L'Hospital's Rule

When first learning about limits, you would often get answers of 0/0. What were your next steps when this happened?

Indeterminate Forms

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L'Hospital's Rule: Suppose f and g are differentiable on an open interval I. If the resulting limit is in the form of an indeterminate form, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

Provided that the limit exist (or is $\pm\infty$). This can also be applied for left- and right-hand limits, as well as limits approaching $\pm\infty$.

Examples

Evaluate the limits:

Ex 1:

$$\lim_{x \to 2} \frac{x^2 - 2x}{x^2 - 6x + 8}$$

Ex 2:

$$\lim_{x \to 4} \frac{x + 3}{x^2 - 3x}$$

Ex 3:

$$\lim_{x \to 0} \frac{1 - \cos(5x)}{x^2}$$

Other Indeterminate Forms

Ex 1:

$$\lim_{x \to 0^+} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$$

Ex 2:

$$\lim_{x \to \infty} x \cdot e^{1/x}$$

Ex 3:

$$\lim_{x \to 0^+} x^x$$